

Basic Introduction To Machine Learning

AL-2002 Artificial Intelligence Spring 2026

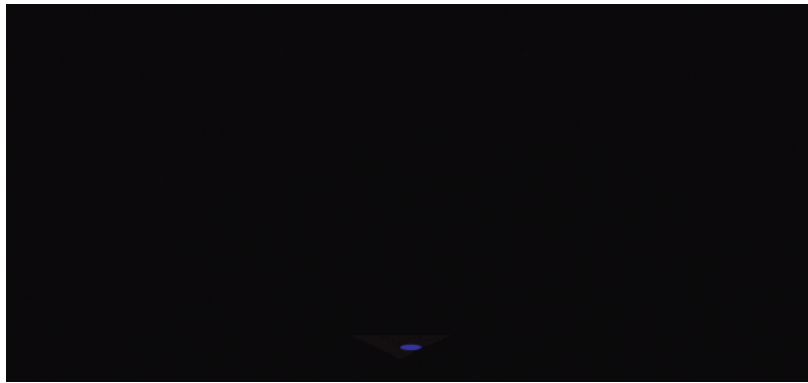
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👋 Let's begin!

🔬 Overview

Machine Learning is a fundamental subfield of artificial intelligence that focuses on analyzing and interpreting patterns and structures in data. It enables reasoning, learning, and decision-making outside of human interaction.

Simply, machine learning allows the user to feed a computer algorithm an immense amount of data and have the computer analyze and make data-driven recommendations and decisions based on only the input data.



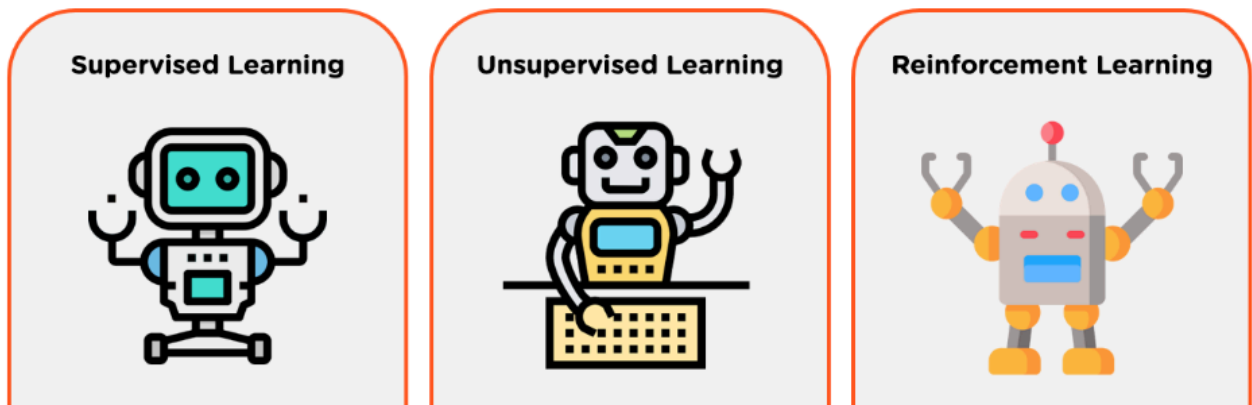
💻 Why it is important?

Machine learning programs aim to automate tasks and draw conclusions from data sets more quickly than humans could by manually analyzing it. It also saves us a lot of time.

Categories in Machine Learning

Machine learning is divided into three different categories:

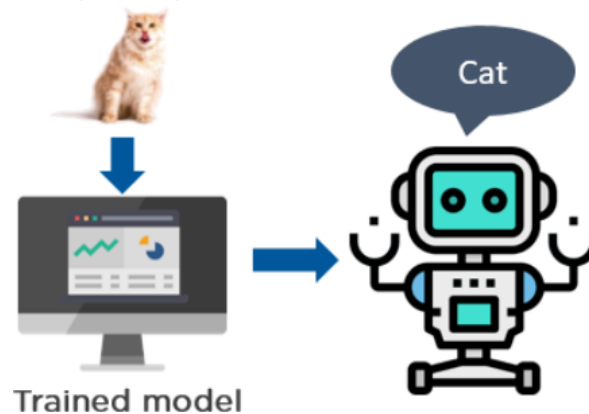
Types of Machine Learning



- Supervised Learning

Supervised learning is a type of machine learning that uses labeled data to train machine learning models. In labeled data, the output is already known.

The model just needs to map the inputs to the respective outputs.



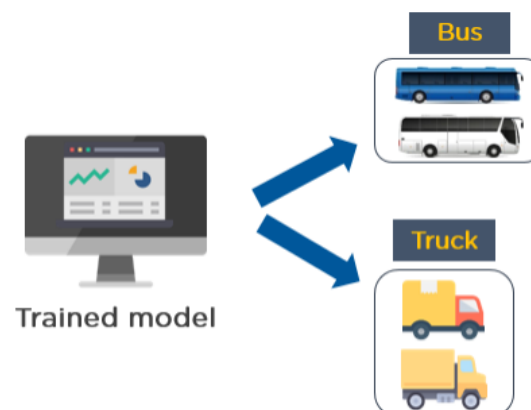
Algorithms:

Some of the most popularly used supervised learning algorithms are:

- Linear Regression
- Logistic Regression
- Support Vector Machine
- K Nearest Neighbor
- Decision Tree
- Random Forest
- Naive Bayes

• Unsupervised Learning

Unsupervised learning algorithm is trained on unlabeled data, and its goal is to discover structures or patterns within the data without having a specific target variable in mind. Common unsupervised learning tasks include dimensionality reduction and clustering.



Algorithms:

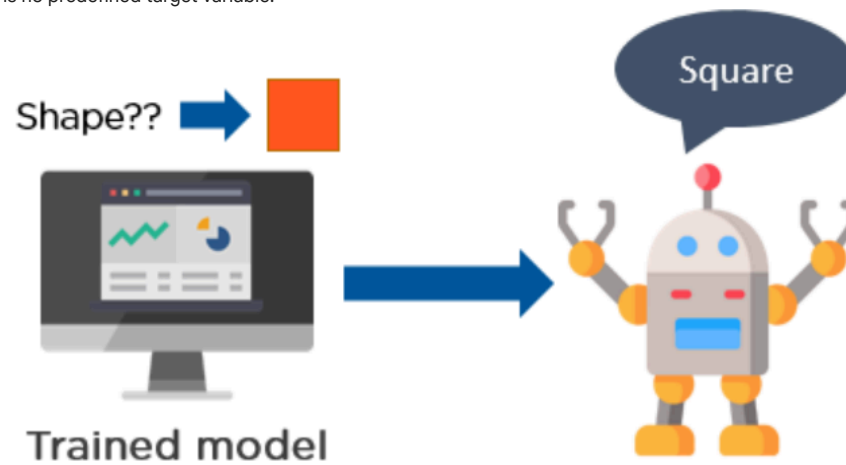
Some of the most popularly used supervised learning algorithms are:

- K Means Clustering
- Hierarchical Clustering
- DBSCAN
- Principal Component Analysis

• Reinforcement Learning

Reinforcement Learning trains a machine to take suitable actions and maximize its rewards in a particular situation. It uses an agent and an environment to produce actions and rewards. The agent has a start and an end state. But, there might be different paths for reaching the end state, like a maze. In this

learning technique, there is no predefined target variable.



Algorithms:

Some of the most popularly used supervised learning algorithms are:

- Q Learning
- Sarsa
- Monte Carlo
- Deep Q network

Basic Structure

The main steps performed on any dataset generally include the following:

1. Data Collection:

2. Data Exploration and Analysis:

- **Exploratory Data Analysis (EDA):** Understand the dataset by summarizing its main characteristics, often using visual methods.
- **Statistical Analysis:** Calculate descriptive statistics (mean, median, mode, standard deviation) to understand the data distribution.

3. Data Cleaning:

- Handling Missing Values.
- Handling Outliers.
- Correcting Data Types.

4. Data Preprocessing:

- Normalization/Standardization.
- Encoding Categorical Variables.
- Feature Selection/Extraction.
- Data Splitting.

5. Model Selection:

- Choosing the Right Model.
- Initialize the chosen models with parameters.

6. Model Training:

- Fit the model to the training data.
- Hyperparameter Tuning.

7. Model Evaluation:

- **Validation:** Evaluate the model on the validation set to tune hyperparameters and prevent overfitting.
- **Testing:** Assess the model performance on the test set using metrics like accuracy, precision, recall, F1-score, RMSE, etc.

8. Model Deployment:

- Integration.
- Monitoring and Maintenance.

9. Model Interpretation:

✖ 🏗️ Import Necessary Libraries

```
import pandas as pd                #For data manipulation and analysis
import numpy as np                 #For numerical operations
import statsmodels.api as sm       #For statistical modelling
import scipy.stats as st           #For statistical functions
import matplotlib.pyplot as plt    #For plotting graph
import seaborn as sn               #For statistical data visualization
from sklearn.metrics import confusion_matrix #Confusion matrix
import matplotlib.mlab as mlab     #For 2D plotting
%matplotlib inline
import seaborn as sns # for statistical data visualization
```

✖ 📦 Load and Transform The Dataset

By focusing on the most relevant and informative features for the analysis we drop some columns initially.

```
dataset=pd.read_csv("Live.csv")
#print(heart_df.head())
```

✖ 🏗️ Exploratory Data Analysis

dataset.head()

	status_id	status_type	status_published	num_reactions	num_comments	num_shares	num_likes
0	246675545449582_1649696485147474	video	4/22/2018 6:00	529	512	262	4
1	246675545449582_1649426988507757	photo	4/21/2018 22:45	150	0	0	1
2	246675545449582_1648730588577397	video	4/21/2018 6:17	227	236	57	2
3	246675545449582_1648576705259452	photo	4/21/2018 2:29	111	0	0	0
4	246675545449582_1645700502213739	photo	4/18/2018 3:22	213	0	0	2

Next steps:

[Generate code with dataset](#)[New interactive sheet](#)

```
# Last rows
dataset.tail()
```

[Show hidden output](#)

```
# Structure
str(dataset)
```

[Show hidden output](#)

```
#Check shape of the dataset
dataset.shape
```

```
(7050, 16)
```

```
#Summary
dataset.info()
```

[Show hidden output](#)

```
#Checking the missing values in the dataset
dataset.isnull().sum()
```

[Show hidden output](#)

```
#Drop redundant Columns
dataset.drop(['Column1', 'Column2', 'Column3', 'Column4'], axis=1, inplace=True)
#dataset.drop(['education', 'ExerciseAngina', 'ST_Slope', 'ChestPainType', 'RestingECG'], axis=1, inplace=True)
```

```
#Statistical summary of numerical variables
dataset.describe()
```

	num_reactions	num_comments	num_shares	num_likes	num_loves	num_wows	num_hahas	num_sads	num_angr
count	7050.000000	7050.000000	7050.000000	7050.000000	7050.000000	7050.000000	7050.000000	7050.000000	7050.000000
mean	230.117163	224.356028	40.022553	215.043121	12.728652	1.289362	0.696454	0.243688	0.1131
std	462.625309	889.636820	131.599965	449.472357	39.972930	8.719650	3.957183	1.597156	0.7268
min	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.0000
25%	17.000000	0.000000	0.000000	17.000000	0.000000	0.000000	0.000000	0.000000	0.0000
50%	59.500000	4.000000	0.000000	58.000000	0.000000	0.000000	0.000000	0.000000	0.0000
75%	219.000000	23.000000	4.000000	184.750000	3.000000	0.000000	0.000000	0.000000	0.0000
max	4710.000000	20990.000000	3424.000000	4710.000000	657.000000	278.000000	157.000000	51.000000	31.0000

```
dataset['status_type'].unique()
```

```
array(['video', 'photo', 'link', 'status'], dtype=object)
```

```
# view how many different types of variables are there
len(dataset['status_type'].unique())
```

```
4
```

```
dataset.drop(['status_id', 'status_published'], axis=1, inplace=True)
```

Visualization

```
# Plotting histograms for each attribute
dataset.hist(bins=30, figsize=(20, 15), color='b', edgecolor='black')
plt.suptitle('Histograms of All Attributes', fontsize=20)
plt.show()
```

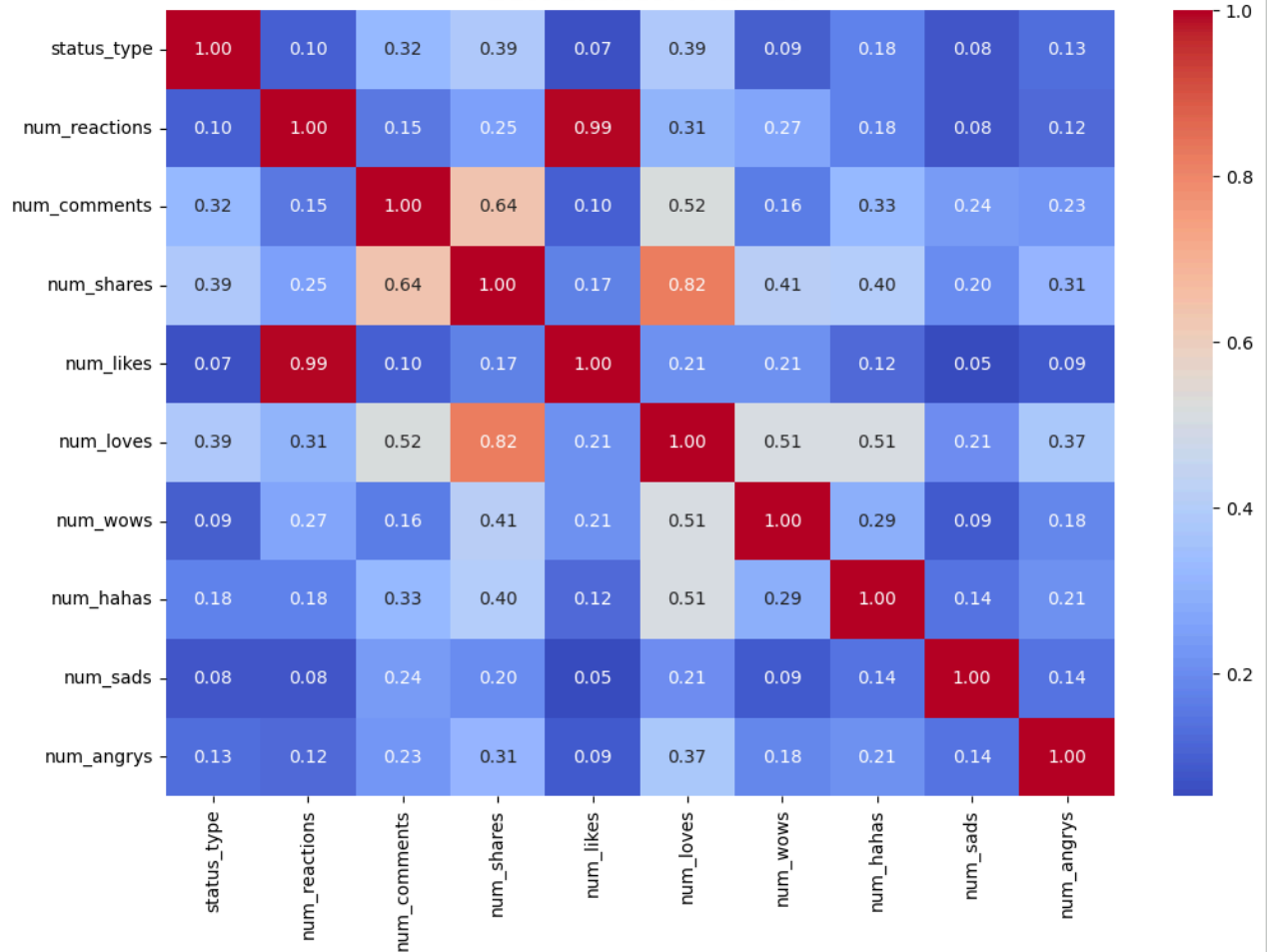
[Show hidden output](#)

Data Preparation

```
# Declare feature vector and target variable
X = dataset
y = dataset['status_type']
```

```
# Correlation matrix
plt.figure(figsize=(12, 8))
correlation_matrix = dataset.corr()
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm', fmt='.2f')
plt.title('Correlation Matrix', fontsize=20)
plt.show()
#corrplot(corr(dataset), type="upper", method="ellipse", tl.cex=0.9)
```

Correlation Matrix



✓ Normalize the Features

Feature Scaling

```
#Converting categorical variables into integer
from sklearn.preprocessing import LabelEncoder
le = LabelEncoder()
X['status_type'] = le.fit_transform(X['status_type'])
# status_type = {'photo': 0, 'video': 1}
y = le.transform(y)
X.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7050 entries, 0 to 7049
Data columns (total 10 columns):
#   Column          Non-Null Count  Dtype
---  -
0    status_type     7050 non-null   int64
1    num_reactions   7050 non-null   int64
2    num_comments    7050 non-null   int64
3    num_shares      7050 non-null   int64
4    num_likes       7050 non-null   int64
5    num_loves       7050 non-null   int64
6    num_wows        7050 non-null   int64
7    num_hahas       7050 non-null   int64
8    num_sads        7050 non-null   int64
9    num_angrys      7050 non-null   int64
dtypes: int64(10)
memory usage: 550.9 KB
```

```
cols = X.columns
from sklearn.preprocessing import MinMaxScaler
ms = MinMaxScaler()
X0 = ms.fit_transform(X)
X = pd.DataFrame(X, columns=[cols])
X.head()
```

	status_type	num_reactions	num_comments	num_shares	num_likes	num_loves	num_wows	num_hahas	num_sads	num_crys
0	1.000000	0.112314	0.024393	0.076519	0.091720	0.140030	0.010791	0.006369	0.019608	0.000000
1	0.333333	0.031847	0.000000	0.000000	0.031847	0.000000	0.000000	0.000000	0.000000	0.000000
2	1.000000	0.048195	0.011243	0.016647	0.043312	0.031963	0.003597	0.006369	0.000000	0.000000
3	0.333333	0.023567	0.000000	0.000000	0.023567	0.000000	0.000000	0.000000	0.000000	0.000000
4	0.333333	0.045223	0.000000	0.000000	0.043312	0.013699	0.000000	0.000000	0.000000	0.000000

Next steps: [Generate code with X](#) [New interactive sheet](#)

✓ Bias in the Model

The bias term (intercept) helps the model capture the overall offset or tendency of the data, providing flexibility to the model to fit the data more accurately.

- Without the bias term, the model might not be able to properly represent the data, leading to underfitting or biased predictions.
- Adding a bias to the model can help reduce error due to overly simplistic assumptions while potentially increasing variance (sensitivity to noise in the data).
- It help to build more generalized models for unseen dataset.

```
# Preprocessing in Statistical modelling
# Adding biasness to the model
from statsmodels.tools import add_constant as add_constant
heart_df_constant = add_constant(heart_df). #Add column of ones
heart_df_constant.head()
```

```
/usr/local/lib/python3.8/dist-packages/statsmodels/tsa/tsatools.py:142: FutureWarning: In a future version of pandas
x = pd.concat(x[::order], 1)
```

	const	Age	Sex	RestingBP	currentSmoker	cigsPerDay	BPMeds	prevalentStroke	prevalentHyp	sysBP	diaBP	BMI
0	1.0	40	0	140	0	0.0	0.0	0	0	106.0	70.0	26.97
1	1.0	49	1	160	0	0.0	0.0	0	0	121.0	81.0	28.73
2	1.0	37	0	130	1	20.0	0.0	0	0	127.5	80.0	25.34
3	1.0	48	1	138	1	30.0	0.0	0	1	150.0	95.0	28.58
4	1.0	54	0	150	1	23.0	0.0	0	0	130.0	84.0	23.10

Hope you liked today's class! Z :)

